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Increasing the wear resistance of hydraulic breaker pick material by cold treatment

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Abstract

It is shown that cold treatment of hydraulic breaker pick materials – steels 38XM, 55C2, У8 and X12МФА (5-hour holding of specimens at −75 °C) – reduces the retained austenite content (by 44 ÷ 88%) and increases the carbide content (by 15 ÷ 98%) in hardened steel, which is accompanied by a slight increase in the strength characteristics ($\sigma_{0.2}$, σ_B) and hardness of the tested materials, with a simultaneous decrease in impact toughness. As a result of exposure to cold, the abrasive wear resistance increases substantially (by 18 ÷ 35%).

Keywords: *hydraulic breaker picks, carbon and alloy steels, cold treatment, increase in wear resistance.*

When excavators with an active-action bucket are operated on the hard and abrasive rocks typical of Vietnam's non-metallic raw-material deposits, the main cause of failure of the working tool – the picks – is their limiting wear (Fig. 1), caused by the insufficient wear resistance of the materials used – high-carbon and alloy steels.

It is known [1, 2, 3] that if the martensite finish point M_k of a steel lies below room temperature, then after quenching retained austenite remains in the structure of the material, and its amount increases with increasing content of carbon and alloying elements in the steel. The presence of retained austenite in the structure of the material of hardened parts adversely affects its physical and mechanical properties and manifests itself, first of all, in reduced hardness and, as a consequence, reduced wear resistance.

[Figure 1 — see original PDF, page 2] Hydraulic breaker pick (ATLAS COPCO HB2500) made of steel 40XH with a worn head part (left) after 58 days of work breaking oversized granite blocks at a quarry in the Binh Dinh area (Vietnam)

Cold treatment, which consists in cooling hardened steel below its M_k temperature, promotes the transformation of retained austenite into martensite and the formation, during subsequent tempering, of a large number of very fine carbide particles, which also contributes to improving the service properties of the steel [8, 13]. For this reason, cold treatment is widely used as an additional operation after quenching to improve the wear resistance of tool steels [1, 4-7, 9-12, 14]. At the same time, no information on its use for improving the service properties of mining tool materials subjected to abrasive wear has been found in the literature. The present work compares the structure, mechanical properties ($\sigma_{0.2}$, σ_B, KCV, HRC) and abrasive wear resistance (R_w) of several steels most

widely used as pick materials (38XM, 55C2, У8 and X12МФА) after the heat treatment usually applied in practice (oil quenching plus low or medium tempering) and after additional cold treatment. In the latter case, the specimens were placed, immediately after quenching, in a cryostat with a cooling liquid at a temperature of −75 °C, where they were held for 5 hours.

Microstructural studies and determination of the carbide and retained austenite contents in the steel were carried out using an Axioplan-2 microscope. Physical and mechanical characteristics were determined on standard five-fold specimens using a Zwick/Roell Z100 tensile testing machine, a Zwick/Roell RKP450 pendulum impact tester and a Zwick/Roell ZHU hardness tester. Wear resistance tests were carried out according to the ASTM G99 method [5] (the “pin-on-disk” method) on cylindrical pin specimens (4 mm in diameter, 14 mm in height), which were worn under a constant load (P = 40 N) at a speed of 0.257 m/s against an electrocorundum disk. From the results of weighing the specimens before and after exposure to the abrasive, the wear resistance was calculated by the formula:

$$W_R = \rho \cdot P \cdot L / m,\ N \cdot m/mm^3, \quad (1)$$

where ρ is the density of the pin material, g/mm^3; m is the mass loss of the pin due to wear, g; L is the wear path (100 m).

The results of the microstructural studies are presented in Figs. 2 and 3 and in Table 1.

[Figure 2 — see original PDF, page 2] Retained austenite content in steel X12МФА without (a) and with (b) cold treatment

[Figure 3 — see original PDF, page 3] Carbide content in steel X12МФА without (a) and with (b) cold treatment

Table 1

Steel grade	Treatment	% carbides		% retained austenite	
		without cold	with cold	without cold	with cold
38XM	Quenching (950 °C, oil) + tempering (200 °C)	0.152÷0.182 0.168	0.281÷0.397 0.333 (+98.2%)	–	–
У8	Quenching (800 °C, water) + tempering (360 °C)	–	–	2.110÷2.314 2.212	1.258÷1.230 1.244 (−43.8%)
X12МФА	Quenching (1000 °C, oil) + tempering (200 °C)	4.429÷5.847 5.616	6.095÷6.986 6.461 (+15.0%)	15.446	1.893 (−87.7%)

Note: the numerator of the fraction (upper line) is the limiting content of the analysed phase; the denominator (lower line) is the arithmetic mean of its 3 values.

The obtained values of the physical and mechanical characteristics of the materials, as the arithmetic mean of the results of 3 experiments, are presented in Table 2.

Table 2

Steel grade	Treatment	Parameter values				
		Yield strength $\sigma_{0.2}$, MPa	Tensile strength σ_B, MPa	Impact toughness KCV, J/cm^2	Hardness, HRC	Wear resistance R_w, N·m/mm^3
38XM	Quenching from 870 °C + tempering (200 °C)	1478	1933	40.6	51.1	69.0
	Quenching from 870 °C + cold + tempering (200 °C)	1529 (+3.5%)	1953 (+1.0%)	33.0 (−18.7%)	51.6 (+1.0%)	81.9 (+18.6%)
	Quenching from 950 °C + tempering (200 °C)	1485	1887	37.5	48.7	51.4
	Quenching from 950 °C + cold + tempering (200 °C)	1486 (+0.1%)	1907 (+1.1%)	33.2 (−11.5%)	49.3 (+1.2%)	59.5 (+15.8%)
55C2	Quenching from 870 °C + tempering (200 °C)	*)	*)	5.6	59.6	97.0
	Quenching from 870 °C + cold + tempering (200 °C)	*)	*)	4.8 (−14.3%)	60.5 (+1.5%)	122.2 (+25.9%)
	Quenching from 870 °C + tempering (360 °C)	*)	*)	7.2	57.8	–
	Quenching from 870 °C + cold + tempering (360 °C)	*)	*)	7.0 (−2.8%)	58.4 (+1.0%)	–
	Quenching from 870 °C + tempering (460 °C)	1496	1617	22.3	46.9	–
	Quenching from 870 °C + cold + tempering (460 °C)	1575 (+5.3%)	1653 (+2.2%)	19.0 (−14.8%)	47.4 (+1.1%)	–

Steel grade	Treatment	Parameter values				
		Yield strength $\sigma_{0.2}$, MPa	Tensile strength σ_B, MPa	Impact toughness KCV, J/cm^2	Hardness, HRC	Wear resistance R_w, N·m/mm^3
У8	Quenching from 870 °C + tempering (200 °C)	*)	*)	5.1	60.4	52.9
	Quenching from 870 °C + cold + tempering (200 °C)	*)	*)	3.9 (−23.5%)	60.9 (+0.8%)	58.5 (+10.6%)
	Quenching from 870 °C + tempering (360 °C)	1677	1810	3.7	53.6 (538)	40.2
	Quenching from 870 °C + cold + tempering (360 °C)	1728 (+3.0%)	1867 (+3.1%)	2.9 (−21.6%)	54.6 (+1.9%)	45.0 (+11.9%)
Х12МФА	Quenching from 870 °C + tempering (200 °C)	*)	*)	3.1	59.8	154.5
	Quenching from 870 °C + cold + tempering (200 °C)	*)	*)	2.5 (−19.4%)	60.4 (+1.0%)	208.2 (+34.8%)

*) – brittle fracture of specimens over a wide range of stresses

As follows from Figs. 2 and 3, as well as Tables 1 and 2, cold treatment, as in [1, 2, 3], leads to a significant decrease in the retained austenite content (by 44 ÷ 88%) and an increase in the carbide content (by 15 ÷ 98%) in hardened steel. This change in structure is accompanied by a slight increase in the strength characteristics ($\sigma_{0.2}$, σ_B) and hardness of the tested materials, with a simultaneous decrease in impact toughness. (The $\sigma_{0.2}$ and σ_B values of steels 55C2 and X12MΦA could not be determined because of the large scatter in the experimental results, caused, as can be concluded, by embrittlement of the specimen material as a result of heat treatment.) Moreover, as the austenitizing temperature for quenching and the tempering temperature decrease, this effect of cold treatment increases. As a result of exposure to cold, the abrasive wear resistance increases by 18 ÷ 35%.

From the research results presented, it can be concluded that cold treatment can be regarded as a promising additional heat-treatment operation for increasing the wear resistance of steels used to manufacture picks of hydraulic breakers working on hard, highly abrasive rocks. In doing so, the most optimal heat-treatment modes should be found at which the negative effect of cold treatment on the impact toughness of the steel will be minimal.

References

1. А. П. Клименко, Холод в машиностроении, М.: Машиностроение, 1969. - 247 с.
2. Е. А. Чудаков, Машиностроение. Энциклопедический справочник. Раздел 2 Материалы машиностроения. Том 3, М.: ГОНТИ, 1947, с. 341.
3. И. И. Новиков, Теория термической обработки металлов, М.: Металлургия, 1978. – 182 с.
4. ASM Handbook. Том. 4 Heat treating, ASM International, 2002, с. 487-495.
5. ASTM G99-2014, ASTM International, 2014.
6. D. Candane, N. Alagumurthi, K. Palaniradja, Effect of cryogenic treatment on microstructure and wear characteristics of AISI M35 HSS, Ж.: International journal of materials science and applications, 2013, с. 56-65.
7. D. N. Collins, Cryogenic treatment of tool steels, Ж.: Advanced materials & processes, 1998, с. 23-29.
8. D. N. Collins, Deep cryogenic treatment of tool steels: A review, Ж.: Heat treatment of metals, 1996.2, с. 40-42.
9. D. N. Collins, J. Dormer, Deep cryogenic treatment of a D2 cold-work tool steel, Ж.: Heat treatment of metals, 1997.3, с. 71-74.
10. Nirmal S. Kalsi; Rakesh Sehgal; Vishal S. Sharma, Cryogenic treatment of tool materials: A review, Ж.: Materials and manufacturing processes- 25, 2010, с. 1077-1100.
11. P. I. Patil, R. G. Tated, Comparison of effects of cryogenic treatment on different types of steels: A review, IJCA Proceedings on international conference in computational intelligence ICCIA(9), 2012, с. 10-29.
12. Pete Paulin, Cold cuts, Ж.: Cutting tool engineering, 1992, с. 61-66.
13. Sub-zero treatment. Technology, processes and equipment, Linde Gas, 2010.
14. T. Yugandhar, P.K. Krishnan, Cryogenic treatment and tts effect on tool steel, 6th International tooling conference, Karlstad university, 2002, с. 671-684.